

Ahlan — Complete Cost Book

Every price, where it goes, measured scenarios & monthly projections

Measured from live production traces — 5 Aug 2026 · FX: 1 USD = 49 EGP

1 · The price sheet — every rate we pay, and to whom

These are the only external costs in the system. Everything else (routing, pricing, bills, automations) is code and costs nothing per use.

What	Provider & model	Rate	Used for
Chat brain (default)	OpenAI gpt-4.1-mini	\$0.40 in / \$1.60 out per 1M tokens	Casual replies, order extraction, phrasing
Premium voice	OpenAI gpt-4.1	\$2.00 / \$8.00 per 1M	Angry guests, VIPs, first-timers (auto-escalate)
Router / choices	OpenAI gpt-4.1-nano	\$0.10 / \$0.40 per 1M	Message classification + option resolving
Voice notes	OpenAI 4o-mini-transcribe	\$0.003 per minute (per second billed)	VN -> text; A/B'd better than whisper-1
Photo understanding	OpenAI gpt-4o-mini vision	\$0.15 / \$0.60 per 1M	Guest sends a photo (complaints, menu asks)
WhatsApp messages	Meta	FREE inside 24h service window	ALL current traffic incl. automations
Template messages	Meta (future, needs approval)	~\$0.03–0.11 per message (Egypt)	Win-backs, reorder reminders — off today
Hosting	Railway (flows + api)	\$5–10 / month flat	The two backend services
Dashboard + DB	Vercel + Supabase	\$0 (free tiers, current scale)	Dashboard hosting, database, storage

Where money does NOT go: no per-order commission, no per-message WhatsApp fee in the 24h window, no per-seat license, no charge for fast-path replies, status pushes, courier pushes, or the recovery/upsell/review automations (all code).

2 · Cost per message type (measured, last 7 days of real traffic)

Message type	What happens inside	Cost (USD)	Cost (EGP)
Fast-path answer	Pure code: hours, location, price, price math, thanks, approved FAQs	\$0.0000	0.00
Router-only turn	nano classify + code dispatch	\$0.0001	0.005
Median chat/order turn	mini extraction + code + mini phrasing	\$0.0012	0.06
Average turn (all real traffic)	mix of the above	\$0.0031	0.15
Heavy chat turn (p90)	long context or premium-model escalation	\$0.0103	0.50
Voice-note add-on (7s)	transcription only, then normal turn	\$0.00035	0.017
Photo from guest	vision classify + reply	+\$0.0002	+0.01
Photo TO guest / menu PDF / receipt	sending media is free (in-window)	\$0.0000	0.00
Automation push	recovery / upsell / review — templated code	\$0.0000	0.00

3 · Real tested scenarios — ran today on production, costs from the traces

Each scenario below was actually executed through the live pipeline today. 'Turns' = guest messages. LLM cost is the trace-recorded total; VN cost added where voice was used.

Scenario (as tested)	Turns	Tokens in/out	LLM	+Voice	TOTAL EGP
"Total for 2 meals and a coke?" (price math)	1	0 / 0	0.000	-	0.00
Hours in Arabic / ingredients / thanks	1	0 / 0	0.000	-	0.00
"Send me a photo of the Iconic" (real image sent)	1	4.8k / 181	0.290	-	0.29
Delivery order, 3 turns, receipt PDF (O-C8GR)	3	5.7k / 312	0.242	-	0.24
Full VOICE order, 3 Arabic VNs (O-7J4F)	3	7.2k / 344	0.235	0.05	0.28
Smart-edit order: absorb+imply+add+remove (O-HBKT)	6	13.5k / 687	0.428	-	0.43
Classic pickup, options step by step (O-6P85)	7	15.9k / 686	0.445	-	0.45
Islam's real buggy VN session (6 VNs, repeats)	9	23.2k / 887	0.488	0.21	0.70
Menu question with PDF attached	1	4.8k / 192	0.501	-	0.50
Franco chit-chat (premium-voice escalation)	1	8.3k / 464	0.760	-	0.76
Injection attempt / bot-or-human / offers honesty	1	~4.5k / 190	~0.45	-	0.45

Read: a COMPLETE order costs 0.24–0.45 EGP. Single chit-chat questions that miss every fast path cost 0.29–0.76 EGP because they carry the menu + persona context; they are the optimization frontier (more approved FAQs in Settings push them to 0.00). The buggy Islam session shows why bugs cost money: 3 wasted turns = +0.2 EGP. Those bugs are fixed.

4 · Scenario walkthroughs — the story, then the math

A. Guest asks price then leaves

"b†CbvE bvDdFHBö/ d 1bvJc& " -> exact DB price. "cFCc "" -> closer.

- Turn 1: fast path (code) = 0.00
- Turn 2: closer (code) = 0.00

Total: 0.00 EGP

B. Classic text pickup order (5 turns)

Item -> options in one block -> answers -> payment -> confirm. Receipt PDF attached.

- 2 order turns @ ~0.06 = 0.12
- options/confirm turns partly code = 0.10
- receipt + PDF send = 0.00

Total: ~0.30 EGP

C. Full VOICE order — 3 Arabic voice notes (measured: O-7J4F)

One VN with everything -> pay VN -> confirm VN. New transcriber.

- 3 VNs x ~7s transcription = 0.05
- 3 LLM turns = 0.23

Total: 0.28 EGP

D. Voice order with accents + one garbled word

"böJb†Dd," gW\$§\$'ÖÖ F6†VB y code (free); truly unclear word -> ONE nano resolver call.

- Same as C = 0.28
- resolver (nano, only when needed) = 0.005

Total: ~0.29 EGP

E. Guest sends a food photo + complaint

Vision reads the photo, bot empathizes, staff flagged if serious.

- vision = 0.01
- reply turn = 0.06-0.50 (mood-dependent)

Total: 0.07-0.51 EGP

F. Abandoned order recovered

Guest builds a draft, goes silent 45 min. Code push with buttons; guest taps Yes; order placed.

- recovery push = 0.00
- "yes" confirm turn = 0.06
- receipt = 0.00

Total: 0.06 EGP -> a recovered ORDER

G. Delivery day for one order

Order (3 turns) + courier auto-assigned + 4 driver-page pushes (on way / near / arrived / delivered) + review ask after 1h.

- order = 0.24
- assignment + 4 pushes + review ask = 0.00

Total: 0.24 EGP for the entire lifecycle

H. Browse-only guest

"bvJdr 'dF(c ,c 'b† " -> list; "photo?" -> image; leaves.

- category ask = 0.43 (LLM lists w/ menu context)
- photo = 0.29

Total: ~0.72 EGP - the most expensive guest type; more FAQs => cheaper

5 · Predictions — per customer, per restaurant, weekly & monthly

Assumptions (conservative): average order = 5 guest turns; 30% of orders fully voice; every order accompanied by 1.5 browse/chat conversations at ~0.30 EGP each; automations free; FX 49.

Volume	Orders cost	Chat/browse cost	AI total /mo	Hosting	ALL-IN /mo	Per week
300 orders/mo	105 EGP	135 EGP	240 EGP	250-490 EGP	490-730 EGP	115-170
1,000 orders/mo	350 EGP	450 EGP	800 EGP	250-490 EGP	1,050-1,290	245-300
3,000 orders/mo	1,050 EGP	1,350 EGP	2,400 EGP	490 EGP	~2,890 EGP	~670
Per-customer view			Value			
One complete order conversation			0.24 - 0.45 EGP			
Typical customer / month (3 orders + 2 browses)			1.5 - 2.0 EGP			

Cost as % of a 300-EGP average ticket	~0.1 - 0.15%
1,000-order restaurant: AI cost per order all-in	~1.05 - 1.29 EGP

6 · Error margins & what could move these numbers

- Everything in section 3 is MEASURED, not estimated; sections 5's mix (turns/order, voice %, browse ratio) is the assumption layer — track it monthly from /api/metrics.
- Upside: every approved FAQ converts a paid chat turn to a free one forever; fast paths already answer common questions at 0.
- Downside risks: premium-voice escalations (angry guests) cost ~4x mini — capped by design to moods/VIPs; very long voice rambles (cap VNs at 60s is recommended); Meta template campaigns when enabled are a separate, per-send marketing budget.
- OpenAI price changes pass straight through; a model-tier switch (Settings -> AI) is one toggle.
- Multi-restaurant: each tenant pays only its own usage; hosting is shared — the marginal restaurant costs ONLY its AI usage (no new fixed cost until scale demands bigger instances).